COMPARATIVE CASE STUDY

ShellHacks winning projects

Evidence from every completed edition 2017 to 2025

Build around a specific task and a demonstrable result. Technology choices vary widely across winners; the evidence does not establish a universal winning stack.

COVERAGE	RESEARCH BASIS
9 completed editions	31 overall podium projects and 5 supplementary category winners
Research cutoff	25 September 2026
Purpose	Help a new team select, scope and demonstrate a competitive ShellHacks project

Prepared by Codex for project planning. Independent research; not an official ShellHacks judging guide.
Adapted from Smartsheet's Case Study Report Template for Microsoft Word.

Executive summary

The strongest commonality is a specific intended task. All 31 overall winners describe one, and 30 describe an interface or physical demonstrator. The tasks range from renting electricity and balancing a ball to learning sign language and monitoring floods. A narrow, understandable result is the most defensible starting point for a new project. [P01–P31]

External services appear in 23 of 31 records, while runtime AI or machine learning is clearly documented in nine. These are descriptive counts, not estimates of how much a feature improves the chance of winning. Undocumented features remain unresolved.

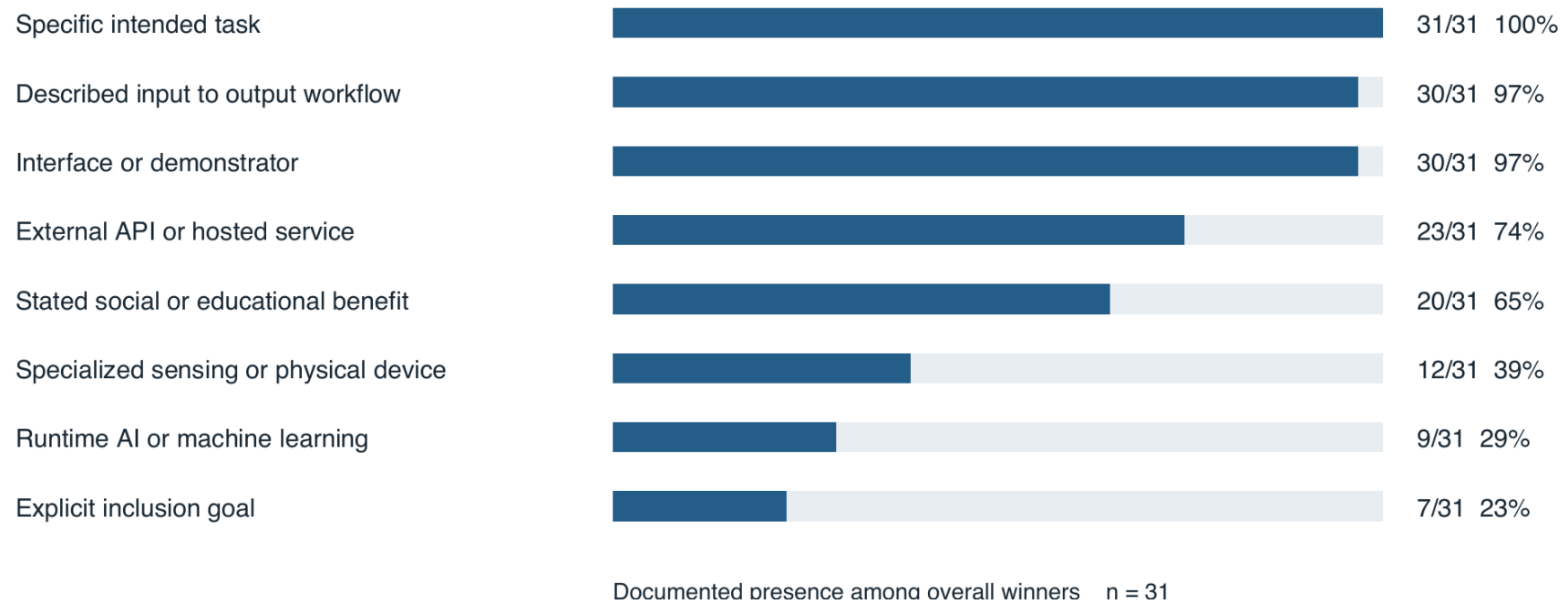


Figure 1. Documented traits in the primary cohort. Each denominator is 31. Categories overlap. Missing, ambiguous, planned and explicitly excluded features are not counted as documented presence. Dataset and source register: P01–P31.

Recommendation. Choose a concrete user task or technical challenge, deliver one inspectable interaction, explain the distinctive implementation, and align with the current rubric and one genuinely relevant prize. Historical winners support this approach; they do not establish that AI, hardware, social impact or a finished commercial product is mandatory.

Introduction and research scope

This study asks what ShellHacks winners have in common and what a new team can reasonably learn from them. It covers all nine completed editions from the event's 2017 launch through 2025. The 2026 event is scheduled for September 25–27 and is incomplete at this cutoff. [E01–E11]

Selection rule. Include every identifiable overall first, second and third place, retain both 2018 third-place awards, and include both 2022 divisions. The 2025 pages share a combined overall award label, so no individual ranks are assigned. Five 2025 category winners form a separate illustrative cohort. This is an overall-podium census under that rule, not a census of all prize-winning projects.

Edition	Sample	Gallery projects	Devpost users	Coverage note
2017	3	38	105	Three ranked overall winners
2018	4	65	158	Two third-place recipients
2019	3	61	186	Three ranked overall winners
2020	3	222	564	Three ranked overall winners
2021	3	265	751	Three ranked overall winners
2022	6	238	676	3 in person + 3 virtual
2023	3	232	798	Three ranked overall winners
2024	3	257	871	Three ranked overall winners
2025	3	245	840	Combined label; ranks unresolved

Table 1. Platform counts observed during research, not audited event attendance or contemporaneous entrant totals. These counts are context only; they are not denominators for winning rates. Event overview and gallery sources E01–E09.

Awards were verified in each project's event-specific Devpost winner block. Feature descriptions come from public project pages. We did not reproduce code, test live applications, interview teams, obtain judging scores or compare a matched set of nonwinners.

Decision criteria and assumptions

The unit of analysis is a unique project within its edition. A project with several awards counts once. We code public documentation rather than assume that a technology tag, a future plan or a working link establishes a finished capability.

Field	Documented presence means	Interpretation boundary
Specific task	A named user need, entertainment experience or technical objective	A broad descriptive property, not a differentiator proven against nonwinners
Workflow	A described input, processing step and resulting output or action	Described behavior may include unfinished implementation
Interface or demonstrator	GUI, terminal, physical demonstrator or other inspectable interaction	Does not rate visual polish or verify live-demo reliability
External service	Named external data/API, hosting, database or tunnelling service	Local SDKs, libraries and private local blockchain APIs do not qualify
Social or educational goal	Explicit health, learning, inclusion, community, civic or environmental purpose	Intended benefit; no independently measured real-world impact
Hardware or sensing	Specialized sensor, dedicated camera, headset or physical device	Includes built-in LiDAR and off-the-shelf devices; not necessarily custom hardware
Runtime AI or ML	Named model or learned component documented as implemented in the prototype	Excludes AI-assisted coding, ambiguous CV and future-only AI
Inclusion	Explicit disability access or an underserved-user objective	General convenience or ordinary usability alone is insufficient

All other codes are “not established.” This combines missing, ambiguous, deferred and explicitly excluded evidence; it never means every uncounted project lacked the feature. Raw notes retain the distinctions. Data preparation reconciled the educational goal of Tango and the local API use in Rent-A-Plug. [P01, P14]

Limits. Winner selection, uneven documentation and later page edits prevent causal inference. Small edition samples and changing sponsors prevent reliable trend or win-probability estimates. A nonwinner comparison and judging scores would be needed to test predictive factors.

Published judging criteria

Judging guidance changes across editions. It provides a stronger basis for preparation than treating a frequently used technology as a requirement. The table records published wording or a concise paraphrase; no scoring weights were recovered.

Edition	Published criteria or overall award description	Planning implication
2017	No substantive scoring rubric in the overview	Use results as examples, not an inferred scoring system
2018	Design, Creativity, Technology	Explain the interaction and technical work
2019	No substantive rubric in the overview	Do not invent historical weights
2020–2021	Overall prize descriptions emphasize creativity, execution and impact	Connect the idea, implementation and intended value
2022	Creativity, Complexity, Impact	Choose a technical challenge that can be explained and demonstrated
2023	Completion, Originality, Learning, Design, Technology	Show what works and what the team learned
2024–2025	Completion, Originality, Design, Technology, Practicality	Make the usable result easy to judge
2026	No current scoring rubric recovered from inspected public pages	Check the current hacker guide and sponsor briefs before committing scope

Table 2. Sources E01–E10. Historical criteria are not represented as current 2026 rules. The 2024 and 2025 overall-prize descriptions also refer to creativity, execution and impact.

Sponsor fit is a separate decision. The 2025 Google Cloud brief specifies agent architecture, Netflix asks for branching interactive entertainment, and State Farm emphasizes insurance education. A11y Tools, Quests and Conflicts VR, and GreenLight illustrate different responses to those briefs. The five supplementary cases are excluded from primary percentages. [E09, P32–P36]

For 2026, the public FAQ describes a 36-hour build window and teams of one to four. Detailed current scoring rules were not recovered. Historical 2025 rules restricted reused projects and required attribution of external code; teams should verify the current version before building. [E09, E10]

Data analysis across editions

Runtime AI is documented in 0 of 10 early-period projects, 5 of 12 middle-period projects and 4 of 9 recent-period projects. This is evidence of presence in the reviewed descriptions, not proof of adoption growth: documentation, tools and the 2022 dual podium change the comparison.

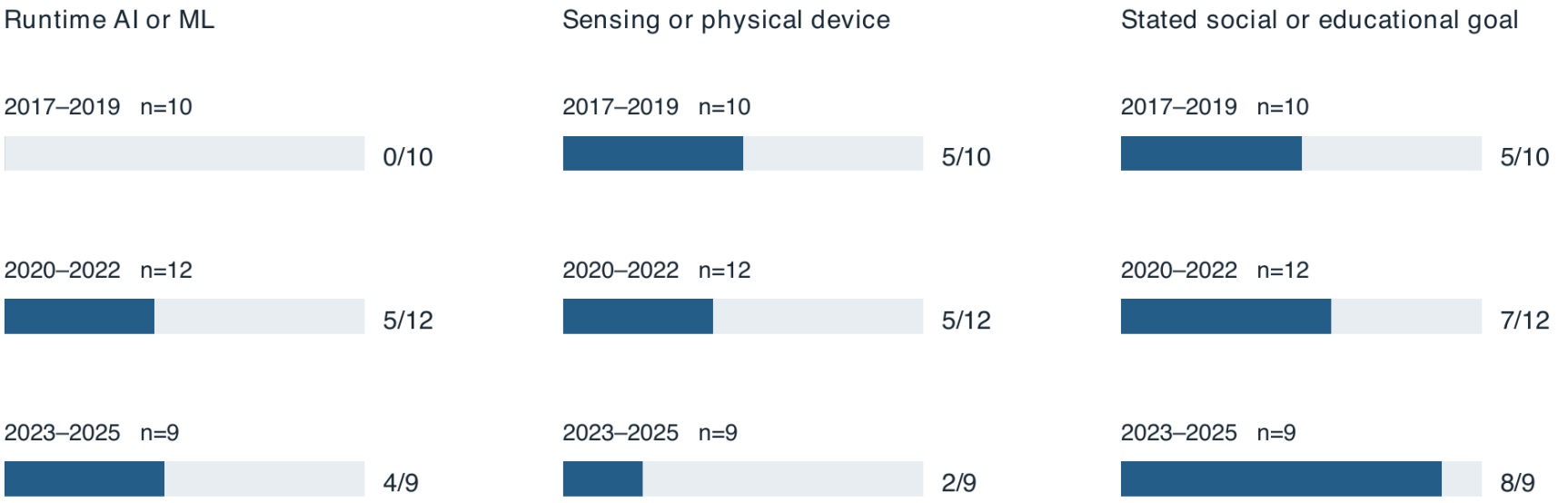


Figure 2. Counts within three edition groups. Bars show documented share on a common 0–100% scale. The unequal denominators include the extra 2018 third-place project and 2022 virtual podium. Sources P01–P31.

What the variation means for a new team

Recent winners include projects with deferred or unclear AI features: ESPER lists future ML work, and EcoSort does not establish a runtime classifier. AI should follow the task and prize brief. [P26, P29]

Hardware appears in 12 of 31 records, from cameras to headsets. Choose it when the interaction benefits and a dependable demonstration is feasible. [P01–P31]

Twenty of 31 state social or educational goals, while The Tilter, Remote Game Sharer and Chess 10 demonstrate other viable directions. Match the impact narrative to the actual project. [P07, P17, P23]

Comparative case analysis

HurriHome connects a local problem to a visible system

The 2018 first-place project links home sensors, cloud storage and a mobile experience for hurricane evacuees. Its authors describe testing wireframes with attendees, then limiting implementation to essential functions. The transferable lesson is to make the main user decision visible and scope deliberately. The page also acknowledges power and connectivity tradeoffs, so an award does not establish deployment readiness. [P04]

The Tilter makes technical execution directly observable

The 2022 in-person first-place project balances a golf ball using camera tracking, control logic and servos. Its result can be inspected immediately. It supports a route based on an engaging engineering challenge without an explicit public-benefit mission. Computer vision alone does not establish a trained model, so it is not counted as documented AI. [P17]

ESPER ties a specific interaction to an accessibility goal

The 2024 first-place project changes web-video brightness and presents video characteristics in a dashboard. Its AI work is described as a future step. This case supports task-specific implementation and accessible design as viable choices without proving medical effectiveness or an AI requirement. [P26]

Lumi Lens separates the concept from demo limitations

A 2025 overall winner, Lumi Lens combines generated dialogue with phone and glasses overlays for neurodivergent users. Live emotion detection encountered blockers and the demonstration used simulation. The lesson is to disclose which components work and which are simulated. The source does not show that judges preferred simulation or that the proposed benefit was validated. [P31]

Common structure with different technology

These cases connect an understandable task to an inspectable behavior. Their implementation choices differ. The conclusion is an evidence-based design direction, not a claim that the same judging rationale was applied to every winner.

Recommendations for a brand new project

Use the following as a preparation framework. Each recommendation is tied to a finding or published criterion; the proposed checks are practical team decisions, not thresholds proven to predict winning.

Order	Build decision	Evidence basis	Concrete self-check
1	Choose one specific task or technical challenge	31/31 overall records have a specific task	A teammate can state who uses it, the input and the useful result
2	Deliver one inspectable core interaction	30/31 describe an interface or demonstrator; completion and design recur in criteria	Demonstrate the main result with one representative input and disclose simulated parts
3	Explain what is distinctive and technically meaningful	Creativity/originality and implementation recur in published criteria	Compare with the nearest existing approach and identify the contribution built during the event
4	Use services or AI only where they help the task	23/31 document services; 9/31 clearly document runtime AI	Name each dependency and explain the user-facing behavior it enables
5	Choose a prize whose brief matches the actual build	Distinct 2025 category winners map to distinct sponsor prompts	Show the required integration or experience in the submitted demo
6	Make claims proportional to evidence	Multiple winners disclose incomplete features; capability reports are untested	Separate demonstrated results, intended benefits and future work

Viable project directions

A focused utility can follow JAM, Freetear or IntelliSchedule; an interactive technical experience can follow The Tilter, Saberfy or Chess 10; an assistive or community workflow can follow Ping Path, ESPER or FloodGuard. These are precedents for choosing a type of problem, not invitations to resubmit or clone existing projects. [P06, P08, P17, P19, P22–P24, P26, P30]

For an AI concept, define the task before selecting a model, show the input and output, and disclose limits. For hardware, identify the smallest stable physical loop. For a sponsor entry, make the qualifying feature central to the experience. The historical evidence does not justify a weighted “winning score.”

Implementation plan and conclusion

This 36-hour schedule translates the recommendations into work. The time allocations are editorial suggestions, not timings measured from winners. Keep all preparation and implementation within the current event rules. [E10]

Window	Deliverable	Reason for the allocation
Hours 0–3	One task, selected prize brief and storyboard	Avoid a broad project whose value cannot be judged quickly
Hours 3–12	Small working input-to-output path	Resolve the central technical risk before expanding scope
Hours 12–24	Integrated core experience and dependency checks	Make the task repeatable; add only features that strengthen the main result
Hours 24–30	Usability review and documented test cases	Check representative inputs and make limitations visible
Hours 30–34	Recorded demonstration, concise project page and source attribution	Present the behavior, implementation and current status clearly
Hours 34–36	Submission validation and rehearsal	Confirm current eligibility, required links and sponsor requirements

Conclusion directly supported by the data

All 31 overall-winning records identify a specific task; 30 describe an interface or demonstrator. Their technology choices and intended audiences vary. A new project should therefore begin with a concrete task, make the central result inspectable, and explain why its implementation is distinctive and useful under the current judging criteria.

The evidence supports choices, not necessary and sufficient conditions for winning. AI, hardware and an explicit social mission are not established as universal requirements. Nor do the records show that all winners shipped a complete product. With no comparable nonwinner cohort or judging scores, there is no defensible probability forecast or guaranteed recipe.

The most useful next step is to select a narrow project that the team can demonstrate credibly, then use the current rubric and sponsor brief to decide what deserves the remaining build time.

Exhibit 1 Project feature matrix

Primary cohort of overall winners. Y means documented presence. A dash means not established and must not be read as a confirmed absence. All records have a specific task; sock.it is the only record without a sufficiently described input-to-output workflow.

ID	Year	Project	Demo	Service	Benefit	Device	AI	Inclusion
P01	2017	Rent-A-Plug	Y	—	—	Y	—	—
P02	2017	Pocket Park	Y	Y	Y	—	—	—
P03	2017	Mic Drop	Y	—	—	—	—	—
P04	2018	HurriHome	Y	Y	Y	Y	—	—
P05	2018	Speak Free	Y	Y	Y	Y	—	Y
P06	2018	JAM	Y	Y	—	—	—	—
P07	2018	Remote Game Sharer	Y	—	—	—	—	—
P08	2019	Freetear	Y	Y	—	—	—	—
P09	2019	No Vote Left Behind	Y	Y	Y	Y	—	Y
P10	2019	sock.it	—	Y	Y	Y	—	—
P11	2020	Farsense	Y	—	Y	Y	—	—
P12	2020	Social Distance Beer Pong	Y	—	Y	Y	—	—
P13	2020	CloudPlate	Y	Y	—	Y	Y	—
P14	2021	Tango	Y	Y	Y	—	Y	—
P15	2021	Bizchain	Y	Y	Y	—	Y	—
P16	2021	Kudos	Y	Y	—	—	—	—

Demo = interface or demonstrator. Service = external API or hosted service. Benefit = stated social or educational goal. Device = specialized sensing or physical hardware. AI = runtime learned component. Full definitions appear in Decision criteria and assumptions. Source IDs resolve in the project catalogue and reference register.

Exhibit 2 Project feature matrix

Primary cohort of overall winners. Y means documented presence. A dash means not established and must not be read as a confirmed absence. All records have a specific task; sock.it is the only record without a sufficiently described input-to-output workflow.

ID	Year	Project	Demo	Service	Benefit	Device	AI	Inclusion
P17	2022	The Tilter	Y	—	—	Y	—	—
P18	2022	Pegasus	Y	Y	—	—	—	—
P19	2022	Saberfy	Y	Y	—	—	—	—
P20	2022	DAOchow	Y	Y	Y	—	Y	Y
P21	2022	AlgoSaurus	Y	Y	Y	—	—	—
P22	2022	Ping Path	Y	Y	Y	Y	Y	Y
P23	2023	Chess 10	Y	—	—	—	—	—
P24	2023	IntelliSchedule	Y	Y	Y	—	—	—
P25	2023	MedVoyage	Y	—	Y	—	—	—
P26	2024	ESPER	Y	Y	Y	—	—	Y
P27	2024	ARSL	Y	Y	Y	Y	Y	Y
P28	2024	Diffusion Earth	Y	Y	Y	—	Y	—
P29	2025	EcoSort	Y	Y	Y	—	—	—
P30	2025	FloodGuard	Y	Y	Y	—	Y	—
P31	2025	Lumi Lens	Y	Y	Y	Y	Y	Y

Demo = interface or demonstrator. Service = external API or hosted service. Benefit = stated social or educational goal. Device = specialized sensing or physical hardware. AI = runtime learned component. Full definitions appear in Decision criteria and assumptions. Source IDs resolve in the project catalogue and reference register.

Exhibit 3 Project feature matrix

Supplementary category winners from 2025. Y means documented presence. A dash means not established and must not be read as a confirmed absence. All five projects describe a specific task and an input-to-output workflow.

ID	Year	Project	Demo	Service	Benefit	Device	AI	Inclusion
P32	2025	A11y Tools	Y	Y	Y	—	Y	Y
P33	2025	Agent Zero	Y	Y	Y	—	Y	Y
P34	2025	GreenLight	Y	Y	Y	—	Y	—
P35	2025	Happy Head	Y	Y	Y	Y	Y	—
P36	2025	Quests and Conflicts VR	Y	—	—	Y	—	—

Demo = interface or demonstrator. Service = external API or hosted service. Benefit = stated social or educational goal. Device = specialized sensing or physical hardware. AI = runtime learned component. Full definitions appear in Decision criteria and assumptions. Source IDs resolve in the project catalogue and reference register.

Supplementary data and sponsor interpretation

Measure	Documented among 5 category winners
Interface or demonstrator	5/5
External API or hosted service	4/5
Stated social or educational benefit	4/5
Specialized sensing or physical device	2/5
Runtime AI or machine learning	4/5
Explicit inclusion goal	2/5

This selection illustrates different prize routes and is not representative of all sponsor winners. It is never pooled into the primary n=31 percentages. Happy Head won a first-time-hacker category; the other four received named sponsor awards. [P32–P36]

Project catalogue 1 to 3

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P01 2017 Rent-A-Plug

Award: 1st Place.

People or machines buying short intervals of electricity. Cryptocurrency payment activates a timed physical power outlet.

Technology: Ethereum, Solidity, Raspberry Pi, JavaScript, relay.

Limitations: Capability descriptions are participant reports and were not independently tested.

Source P01 https://devpost.com/software/rent-a-plug

P02 2017 Pocket Park

Award: 2nd Place; The ShellHack.

Miami drivers and owners of unused parking. Hosts list spaces and prices; drivers choose a map location and reserve parking.

Technology: React, Redux, Node.js, Express, MongoDB, Google Maps.

Limitations: Capability descriptions are participant reports and were not independently tested.

Source P02 https://devpost.com/software/pocket-park

P03 2017 Mic Drop

Award: 3rd Place.

Presenters sharing a venue speaker system. An Android phone streams microphone audio through a socket to a host computer.

Technology: Java, Android Studio, sockets.

Limitations: Capability descriptions are participant reports and were not independently tested.

Source P03 https://devpost.com/software/mic-drop-8a41o3

Project catalogue 4 to 6

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P04 2018 HurriHome

Award: First Place; GE Digital Web Optimization Challenge.

Evacuated homeowners and disaster response groups. Home sensors send status to a cloud platform, mobile app and neighborhood map.

Technology: Arduino, Ionic, Python, Firebase, Heroku, Wix, Google Maps.

Limitations: Capability descriptions are participant reports and were not independently tested.

Source P04 https://devpost.com/software/hurrihome

P05 2018 Speak Free

Award: Second Place.

People practising public speaking. A virtual audience reacts while the user rehearses a speech.

Technology: Unity, C#, Wix.

Limitations: Capability descriptions are participant reports and were not independently tested.

Source P05 https://devpost.com/software/speak-free

P06 2018 JAM

Award: Third Place.

Job seekers tracking multiple applications. Dashboard compares application stage, location and compensation.

Technology: Wix database, Wix Code, React, JavaScript.

Limitations: Capability descriptions are participant reports and were not independently tested.

Source P06 https://devpost.com/software/jam-83i15d

Project catalogue 7 to 9

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P07 2018 Remote Game Sharer

Award: Third Place.

Players sharing access to games remotely. Client inputs control an emulated controller on a host device.

Technology: C, C++, JavaScript, Node.js.

Limitations: Capability descriptions are participant reports and were not independently tested.

Source P07 https://devpost.com/software/sharerhost

P08 2019 Freetear

Award: First Place Overall.

Developers lose time configuring deployment, databases and hosting for side projects. CLI templates an application and automates connections to free-tier deployment services.

Technology: Node.js, Python, Flask, React, shell, Zeit.

Limitations: No measured deployment-time improvement or independent validation.

Source P08 https://devpost.com/software/freetear

P09 2019 No Vote Left Behind

Award: Second Place Overall; JP Morgan Chase & Co. - Best Hack for Social Good; Citrix - Most Inclusive Hack.

People with limited time or resources face voter-registration barriers. Raspberry Pi generates QR/RFID credentials; a web form fills an official registration PDF and sends it to the relevant jurisdiction.

Technology: Raspberry Pi, RFID/NFC, QR, Azure VM, Flask, MySQL, Python, Interfax, reCAPTCHA.

Limitations: No independent election-process, security or real registration validation; do not adopt team's security claims as verified.

Source P09 https://devpost.com/software/no-vote-left-behind

Project catalogue 10 to 12

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P10 2019 sock.it

Award: Third Place Overall; ShellHacks - Best Hardware Hack.

Caregivers need information about an infant's heart rate, oxygen levels and sleep. Senses heart rate and oxygen saturation; the public description stops before explaining the subsequent data flow.

Technology: Arduino, Node.js, Pusher, React, sockets.

Limitations: Most narrative sections empty. No clinical accuracy, completed UI or alerting workflow established.

Source P10 https://devpost.com/software/sock-it

P11 2020 Farsense

Award: Best Overall - 1st Place; Wolfram Award by Wolfram Language.

Remote environmental monitoring can help assess disaster-related property and environmental damage. Battery-powered sensor stations send weather observations over a LoRa mesh to a dashboard.

Technology: ESP32, RFM95, MTK3333 GPS, BME680, LoRa, Arduino, C, React.

Limitations: Damage prediction is a potential use; an operational predictive model or disaster-outcome improvement is not established.

Source P11 https://devpost.com/software/farsense

P12 2020 Social Distance Beer Pong

Award: Best Overall - 2nd Place; Wolfram Award by Wolfram Language.

Friends wanted to play a familiar social game while remaining physically distant during COVID-19. Infrared cup sensors and a microcontroller send data by USB to a C++ terminal game.

Technology: Adafruit ItsyBitsy, Arduino, IR sensors, C++, 3D printing.

Limitations: No remote network architecture or measured disease-risk reduction documented; terminal interface counts as interaction, not polished graphical UI.

Source P12 https://devpost.com/software/social-distance-beer-pong-8nkfe6

Project catalogue 13 to 15

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P13 2020 CloudPlate

Award: Best Overall - 3rd Place; Best Potential Startup by 1517 Fund (Undergraduate students only); Wolfram Award by Wolfram Language.

Dealership arrival and service check-in can be slow and impersonal. Camera/OCR identifies an arriving car; dashboard displays customer and service records; personalized SMS greets the customer.

Technology: Google Cloud Vision OCR, Google Cloud Datastore, Cloud Storage, Twilio, Python, Flask, HTML/CSS/JavaScript.

Limitations: No quantified time savings or accuracy benchmark; repository kept private, code submitted as archive.

Source P13 https://devpost.com/software/cloudplate

P14 2021 Tango

Award: Best Overall - 1st Place.

At-home dance learners lack interactive feedback from one-way tutorials. Pose estimation compares a learner to a reference routine; timing and body-size adjustments support live annotations, charts and replay.

Technology: MediaPipe, OpenCV, dynamic time warping, React, Chakra UI, Recharts, FastAPI, FFmpeg, SQLite, ngrok.

Limitations: No validated coaching efficacy; responsive layout and robust edge-case handling are future work. Ordinary video input is not counted as custom hardware.

Source P14 https://devpost.com/software/tango-dikvm2

P15 2021 Bizchain

Award: Best Overall - 2nd Place; Best Use of Notifications by Courier; CryptoCurrency Portfolio/Trading Application by CoinRoutes Inc.

Consumers and local merchants face complex cryptocurrency payments and portfolio management. Shop at local merchants, manage crypto holdings, run trading strategies and receive price alerts.

Technology: React, Flask, Google App Engine, AutoML Tables, Firestore, CoinRoutes, CryptoCompare, Courier, Twilio, Figma.

Limitations: No investment-return or model-accuracy validation; shopping/portfolio integration remained an improvement area. Financial simplicity is not coded as disability accessibility.

Source P15 https://devpost.com/software/we-can-change-this-at-any-time

Project catalogue 16 to 18

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P16 2021 Kudos

Award: Best Overall - 3rd Place; Best Use of Google Cloud by Google.

Students and other users struggled to maintain productivity during the pandemic. Collaborate on tasks/projects and earn completion rewards; public page primarily documents design and intended app behavior.

Technology: React Native, JavaScript, Firebase, Figma.

Limitations: Explicit future work includes fixing authentication, continuing mobile-app construction and implementing reward finances. This is a counterexample to claims that all winners had a complete working product.

Source P16 https://devpost.com/software/kudos

P17 2022 The Tilter

Award: First Place.

Division: in person.

A concrete control-engineering challenge: keep a golf ball balanced on a moving platform. Phone camera tracks the ball; Python PID control sends serial commands to Arduino servos.

Technology: Python, PID control, C++, Arduino, DroidCam, servo motors.

Limitations: Engineering demonstrator; no explicit civic/health/education beneficiary is claimed. Future joystick game is a proposal.

Source P17 https://devpost.com/software/the-tilter

P18 2022 Pegasus

Award: First Place (Virtual); MITRE: Building Web3 (4th Place).

Division: virtual.

Help organizations search securities data and identify source-code vulnerabilities. Upload CSV data for indexed queries or source code for rule-based vulnerability scanning and alerts.

Technology: React, Node.js, Python, Flask, Regex, IPFS, AWS Lambda, AWS API Gateway.

Limitations: Authors explicitly linked ideation to Schonfeld and MITRE challenges. Scanning is custom regex-based; do not relabel it ML.

Source P18 https://devpost.com/software/pegasus-5x03j1

Project catalogue 19 to 21

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P19 2022 Saberfy

Award: Second Place; ShellHacks - Best First-Time Hacker.

Division: in person.

Beat Saber players lack maps for some preferred songs; manual creation takes time. Supply a Spotify song link; Python translates song characteristics into importable Beat Saber map files with difficulty variants.

Technology: Python, Spotify Web API, spotdl, JSON, Wix.

Limitations: A team member says the direct website-to-server integration was unfinished. VR game output does not establish custom hardware development.

Source P19 https://devpost.com/software/saberfy

P20 2022 DAOchow

Award: Second Place (Virtual); MITRE: Building Web3 (2nd Place).

Division: virtual.

Food-insecure residents need access to community fridges and transparent funding. A decentralized web app connects users to food access and manages funds through a DAO.

Technology: React, Tailwind, Azure, TensorFlow, Aragon, Twilio.

Limitations: ML task is unspecified. USDA-based placement of new fridges is future work, not a demonstrated function.

Source P20 https://devpost.com/software/daochow

P21 2022 AlgoSaurus

Award: Third Place.

Division: in person.

Beginning programmers struggle to understand sorting algorithms. Choose an algorithm, observe animated sorting, and adjust chart size or speed while reading explanations.

Technology: TypeScript, React, Firebase.

Limitations: Authentication, tutoring and chat are proposed future additions. Ordinary beginner education is not automatically disability/inclusion evidence.

Source P21 https://devpost.com/software/algosaurus

Project catalogue 22 to 24

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P22 2022 Ping Path

Award: Third Place (Virtual).

Division: virtual.

People with visual impairments need safer, more independent navigation around obstacles and destinations. LiDAR and ML interpret surroundings; spatial audio guides users, with shared maps and voice interactions.

Technology: Swift, SwiftUI, RealityKit, RoomPlan, CoreML, Firebase Cloud Storage, LiDAR.

Limitations: Authors report conversations with visually impaired users; efficacy not independently validated. Non-LiDAR support is inconsistently described as current versus after the hackathon.

Source P22 https://devpost.com/software/ping-path

P23 2023 Chess 10

Award: Best Overall.

Give chess players a new strategic experience through altered rules and multiple participants. Play a 3D chess variant extending en passant mechanics to most pieces.

Technology: Three.js, Vanta.js, Blender, TypeScript, React.

Limitations: No AI, social-impact or API integration is reported. New mechanics and implementation are described; originality is not independently assessed.

Source P23 https://devpost.com/software/chess-10

P24 2023 IntelliSchedule

Award: Google; Second Place Overall.

College students waste time switching between course selection and professor reviews. Chrome extension places ratings, difficulty, return-interest scores and review summaries alongside FIU course choices.

Technology: Python, Beautiful Soup, JavaScript, HTML, CSS, GraphQL data retrieval.

Limitations: No sponsor rank stated on award label. AI classification remains ambiguous; do not count proposed chatbot as implemented.

Source P24 https://devpost.com/software/intellischedule

Project catalogue 25 to 27

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P25 2023 MedVoyage

Award: Third Place Overall.

Offer users a guided route from symptom questions to a preliminary diagnostic suggestion. A website walks users through a JavaScript binary decision tree.

Technology: HTML5, CSS3, JavaScript, binary decision tree.

Limitations: A binary decision tree is not necessarily trained ML. Accuracy and safety claims are unvalidated; support for different screen sizes is explicitly future work.

Source P25 https://devpost.com/software/medvoyage

P26 2024 ESPER

Award: Best Overall; Assurant.

Reduce abrupt brightness changes in web video for people with epilepsy. A browser extension measures frame luminosity, applies CSS brightness adjustments, and uploads video characteristics to an analytical dashboard.

Technology: JavaScript, Chrome Extension API, YouTube API, Java, Vert.x, Snowflake, Snowpark, Streamlit, DigitalOcean.

Limitations: No clinical outcome is demonstrated. Sponsor label has no rank.

Source P26 https://devpost.com/software/esper-fy6xeh

P27 2024 ARSL

Award: Second Best Overall; Microsoft; Google.

Make ASL alphabet learning interactive and connected to surrounding objects. Quest passthrough identifies objects; hand tracking and a vibrating glove guide the learner through spelling.

Technology: Unity, Meta XR, Python, PyTorch, OpenCV, TCP, ngrok, C, C#, haptic glove.

Limitations: Object/gesture recognition accuracy needs improvement; whole-word mode is future work. Sponsor ranks unspecified.

Source P27 https://devpost.com/software/arsl-vr-asl-learning

Project catalogue 28 to 30

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P28 2024 Diffusion Earth

Award: Third Best Overall.

Let learners explore environments beyond static images. Start with an image, prompt or address; generate depth/point clouds and synthesize new views as the user moves.

Technology: Python, Open3D, Stable Diffusion, Marigold, Moondream, Creative Upscaler, Next.js, Tailwind, Google Maps Street View API.

Limitations: Authors report reducing frame generation from about 120 seconds to 10–15 seconds; this is self-reported. Urban planning/conservation applications are future proposals.

Source P28 https://devpost.com/software/diffusion-earth

P29 2025 EcoSort

Award: 1st, 2nd, & 3rd Best Overall; Wix.com & Base44.

Students and community members uncertain about correct recycling and donation. Photograph an item, receive disposal guidance, review scans and find donation or drop-off locations.

Technology: Base44, JavaScript, CSS, GitHub.

Limitations: Hardware and some mapping ambitions were deferred; effectiveness was not independently measured.

Source P29 https://devpost.com/software/ecosort-ofyuie

P30 2025 FloodGuard

Award: 1st, 2nd, & 3rd Best Overall.

Residents, emergency responders and city officials need earlier flood warnings. Parallel agents monitor weather, infrastructure and alerts, then coordinate risk notifications.

Technology: Google ADK, A2A, Gemini, Google Cloud, Prisma, Neon, React, TypeScript.

Limitations: Direct city-system connections, SMS/push delivery and wider model training are future work; forecast accuracy is unverified.

Source P30 https://devpost.com/software/floodguard-okgzty

Project catalogue 31 to 33

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P31 2025 Lumi Lens

Award: 1st, 2nd, & 3rd Best Overall.

Neurodivergent users navigating emotionally difficult social situations. An emotion input drives generated supportive dialogue and playful phone or glasses overlays.

Technology: Gemini API, Swift, Python, Raspberry Pi, DeepFace, OLED display.

Limitations: Live emotion detection encountered blockers; the demo used simulation. User studies are future work, so benefit is an intended outcome.

Source P31 https://devpost.com/software/magical-girl-os

P32 2025 A11y Tools

Award: Google Cloud.

People with mobility, visual or cognitive impairments face barriers on ordinary web pages. Choose extension tools for voice interaction, image description, text simplification, search or font changes.

Technology: Chrome extension, Google ADK, Gemini, FastAPI, Python, JavaScript, Web Audio API.

Limitations: Quota failures affected testing; Chrome speech fallback exists; search navigation remained incomplete.

Source P32 https://devpost.com/software/a11y-b8xg5i

P33 2025 Agent Zero

Award: Capital One.

Individuals and small businesses need help identifying suspicious invoices. Route invoice checks to specialized agents and aggregate risk scores into a readable assessment.

Technology: React, TypeScript, FastAPI, BigQuery, Gemini, TensorFlow, YOLOv11, PyArrow.

Limitations: Accuracy and speed improvement claims lack public evaluation numbers on this page; financial protection is an intended benefit.

Source P33 https://devpost.com/software/agentzero-tnh9p3

Project catalogue 34 to 36

Awards are event-specific Devpost labels. Descriptions are participant-reported; links and prototypes were not independently tested. Feature codes are in Exhibits 1–3. P identifiers also serve as citations.

P34 2025 GreenLight

Award: State Farm.

Drivers need actionable learning from unsafe driving incidents. Upload footage, analyze an incident, display a 3D reconstruction and answer tailored safety quizzes.

Technology: Gemini, Google ADK, Three.js, React, Python, FastAPI, Firebase, Figma.

Limitations: Real-time dashcam analysis was deferred; generated output reliability required prompt refinement. Safety effectiveness is not independently demonstrated.

Source P34 https://devpost.com/software/greenlight-idnv6o

P35 2025 Happy Head

Award: Best First-Time Hacker.

Coaches, athletes and medical staff may miss potentially harmful sporting impacts. Sense acceleration and rotation, trigger display/audio alerts and log events for web review.

Technology: TI LaunchPad, BMI160 IMU, LCD, MediaPipe FaceMesh, OpenCV, Next.js, Supabase, Prisma.

Limitations: A monitoring prototype, not a validated diagnostic device; wireless hardware and correlation with diagnosed concussions are future work.

Source P35 https://devpost.com/software/happy-head

P36 2025 Quests and Conflicts VR

Award: Netflix.

Players seeking branching, immersive tabletop-inspired entertainment. Interact with a VR narrative whose paths respond to player choices.

Technology: Unity, Blender, Mixamo, C#, Quest 3, VR.

Limitations: Team describes an MVP; store publication remained a future plan. This is a sponsor-winning entertainment workflow without documented runtime AI or a social-impact aim.

Source P36 https://devpost.com/software/quests-and-conflicts-vr

References and provenance

Project sources P01–P36 are linked directly in the catalogue. All are public Devpost project pages, accessed 25 September 2026. Awards are verified from the event-specific winner labels; capabilities remain team descriptions. No participant contact details or private data are included.

E01 ShellHacks 2017 overview and gallery Project gallery

E02 ShellHacks 2018 overview and gallery Project gallery

E03 ShellHacks 2019 overview and gallery Project gallery

E04 ShellHacks 2020 overview and gallery Project gallery

E05 ShellHacks 2021 overview and gallery Project gallery

E06 ShellHacks 2022 overview and gallery Project gallery

E07 ShellHacks 2023 overview and gallery Project gallery

E08 ShellHacks 2024 overview and gallery Project gallery

E09 ShellHacks 2025 overview and rules Project gallery 2025 rules

E10 ShellHacks 2026 public site and FAQ

E11 FIU News 3 October 2025 How FIU students run Florida’s largest hackathon

E12 FIU calendar ShellHacks 2026

E13 MLH season 2027 events calendar

Template and data companion

Smartsheet Case Study Report Template for Microsoft Word. Downloaded blank template asset dated March 2024; adapted for research with its landscape page geometry, cover illustration and report structure. Typography, tables, charts and page navigation were revised for this study. Smartsheet is not the author or sponsor of the analysis.

The companion workbook contains the project-level dataset, codebook, formula-based counts and edition context. Its research notes preserve evidence and uncertainty. Raw structured notes and dataset.json permit audit and recalculation. Edition registration and advertised prize figures are context only and do not establish attendance, actual payouts or winning odds.